ABSTRACT

This mini-project focuses on addressing an important engineering problem related to **battery thermal management systems (BTMS) for lithium-ion energy storage applications, including stationary battery storage systems and electric mobility platforms. It focuses on maintaining safe operating temperatures to enhance battery performance, reliability, and lifespan**. The objective of the work is to design and develop **Real-Time Embedded Control Platform For Hybrid Air-PCM Thermal Management in Battery Energy Storage Systems** that effectively overcomes the limitations of existing approaches. The proposed work emphasizes **the primary motivation of this work is to improve the thermal safety, efficiency, and reliability of lithium-ion battery packs by preventing overheating, thermal runaway, and uneven temperature distribution. Additionally, the mini-project aims to achieve automated, cost-effective, and energy-efficient cooling with minimal manual intervention**. A systematic methodology is adopted to analyze the problem, define suitable performance criteria, and develop an optimized solution using established engineering principles.

The methodology adopted in this project involves **an embedded control platform based on ESP32 is integrated with temperature, voltage, and current sensors to enable real-time monitoring and control. A hybrid cooling system using Phase Change Material (PCM) and Peltier-based forced air cooling is implemented with threshold-based digital control, and the design is suitable for further validation using MATLAB/Simulink.** . The system is modeled, implemented, and tested under various operating conditions to evaluate its behavior and performance. Key parameters such as **temperature reduction, thermal uniformity, cooling response time, energy efficiency, power consumption of cooling modules, system stability, and robustness under varying load conditions. Experimental observations emphasize reduction in peak temperature, improved thermal balance, and safe operating limits** are analyzed to validate the effectiveness of the proposed approach. The obtained results demonstrate that the developed system performs consistently and reliably when compared with conventional methods.

The outcomes of this project highlight its potential applicability in **the hybrid Air–PCM embedded thermal management system is applicable to electric and hybrid electric vehicles, where battery safety and longevity are critical, as well as renewable energy systems such as stationary battery energy storage integrated with solar and wind power. It is also relevant to smart grids, industrial automation, robotics, and IoT-enabled energy management systems requiring reliable thermal regulation and real-time monitoring.** . The proposed solution contributes to improved system performance and practical feasibility, making it suitable for real-world implementation. Furthermore, the scope of the work can be extended by incorporating **Future enhancements include advanced control algorithms, machine learning–based thermal prediction, real-time cooling optimization, scalability to multi-pack systems, and integration with IoT, cloud platforms, and AI-assisted BMS or digital twin frameworks.** , thereby enhancing the overall effectiveness of the system.

Chapter 1 Introduction

1.1 Background and Motivation

Battery Energy Storage Systems (BESS) are increasingly used in electric vehicles, renewable energy integration, and stationary power applications, where safe and reliable operation is critical. Lithium-ion batteries generate significant heat during charge–discharge cycles, and inadequate thermal management can lead to overheating, non-uniform temperature distribution, accelerated aging, and safety hazards such as thermal runaway. Conventional air-cooling methods are simple and economical but often fail to handle transient thermal loads and maintain uniform temperatures under demanding operating conditions.

To address these limitations, hybrid thermal management approaches that combine passive and active cooling have gained attention. Phase Change Materials (PCM) can absorb excess heat through latent heat storage and reduce peak temperatures, while controlled airflow helps remove stored heat and restore PCM effectiveness. The motivation of this project is to develop a low-cost, embedded, and intelligent thermal management platform that integrates hybrid Air–PCM cooling with real-time sensing and control. Such a system aims to enhance battery safety, efficiency, reliability, and lifespan while enabling automated, energy-efficient operation suitable for modern BESS applications.

1.2 Thermal Challenges in Battery Energy Storage Systems

Battery Energy Storage Systems (BESS) experience continuous heat generation due to internal electrochemical reactions during charging and discharging. Variations in load demand, ambient temperature, and cell characteristics often lead to uneven heat distribution within battery modules. These non-uniform temperatures result in localized hot spots, which accelerate cell degradation, reduce capacity, and compromise overall system reliability.

Inadequate thermal management can also increase the risk of safety issues such as thermal runaway and premature battery failure. Conventional cooling techniques, particularly air-only cooling, struggle to limit peak temperatures during transient operating conditions and high-power operation. Therefore, addressing thermal challenges such as peak temperature control, temperature uniformity, and energy-efficient heat removal is essential for ensuring the safe, reliable, and long-term operation of battery energy storage systems.

1.3 Hybrid Air–PCM Thermal Management Concept

The hybrid Air–PCM thermal management concept combines passive and active cooling techniques to effectively manage heat generation in battery energy storage systems. Phase Change Materials (PCM) absorb excess thermal energy through latent heat during phase transition, which helps in limiting peak temperatures and reducing temperature gradients among battery cells. This passive buffering capability is particularly effective during transient load conditions and sudden temperature rises.

However, PCM alone can become saturated and lose effectiveness if the stored heat is not removed. To overcome this limitation, controlled forced-air cooling is integrated with the PCM layer. The airflow removes accumulated heat from the PCM and surrounding battery surfaces, restoring the PCM's thermal storage capacity and maintaining steady operating conditions. This combined approach provides improved thermal stability, enhanced safety, and energy-efficient temperature regulation compared to standalone cooling methods.

1.4 Role of Embedded Control Systems

Embedded control systems play a crucial role in ensuring effective, reliable, and energy-efficient thermal management in Battery Energy Storage Systems (BESS). As battery packs operate under varying load and environmental conditions, real-time monitoring and dynamic control are essential to prevent overheating and maintain safe operating temperatures. Embedded controllers enable continuous acquisition of temperature, voltage, and current data from multiple points within the battery module, providing accurate insight into the system's thermal and electrical state.

By integrating sensors, microcontrollers, and power electronic interfaces, embedded systems allow closed-loop control of cooling mechanisms such as fans, pumps, or thermoelectric modules. Control algorithms implemented in firmware—such as threshold-based logic, hysteresis control, or PID regulation—adjust cooling intensity based on real-time measurements, ensuring stable operation without unnecessary energy consumption. This automated control eliminates the need for manual intervention and reduces the risk of human error during operation.

Embedded platforms also support data logging, fault detection, and diagnostic capabilities, which are vital for performance evaluation and system safety. Features such as over-temperature protection, sensor plausibility checks, watchdog timers, and fail-safe actuation enhance reliability and protect the battery system from abnormal conditions. Additionally, embedded systems provide interfaces for human–machine interaction (HMI), allowing users to monitor system status, configure setpoints, and respond to alarms.

Furthermore, the modular and programmable nature of embedded control systems makes them highly scalable and adaptable. They can be extended to multi-pack configurations, integrated with Battery Management Systems (BMS), or connected to IoT and cloud platforms for remote monitoring and predictive maintenance. Overall, embedded control systems form the intelligence backbone of modern hybrid thermal management solutions, enabling safe, efficient, and intelligent operation of battery energy storage systems.

1.5 Chapter Organization

This report is organized as follows: Chapter 2 presents a comprehensive literature review on battery thermal management techniques. Chapter 3 details the methodology and system design of the proposed hybrid Air–PCM platform. Chapter 4 discusses experimental results and performance evaluation. Chapter 5 concludes the work and outlines future enhancements.

Chapter 2 Literature Review

2.1 Overview of Battery Thermal Management

Battery Thermal Management (BTM) is a critical aspect of ensuring the safe, reliable, and efficient operation of lithium-ion energy storage systems. As batteries undergo charging and discharging cycles, electrochemical reactions generate heat, which, if not managed properly, can lead to excessive temperature rise, uneven heat distribution, accelerated aging, and even catastrophic failure such as thermal runaway. Effective thermal management is therefore essential to maintain optimal battery performance, extend lifespan, and enhance safety across various applications including electric vehicles, stationary energy storage, and renewable energy systems.

The primary objectives of battery thermal management systems (BTMS) are to maintain the cell temperature within a desired operating window, minimize temperature gradients between cells, and ensure rapid dissipation of heat during high-load or transient conditions. Various strategies have been developed, ranging from passive approaches such as natural convection and Phase Change Materials (PCM) to active cooling techniques like forced air, liquid cooling, and thermoelectric modules. Each method offers distinct advantages and limitations, influencing the choice of cooling strategy based on application requirements, energy efficiency, and system cost.

Modern BTMS also integrate sensing, control, and data acquisition systems to actively monitor and regulate thermal conditions. Sensors measure cell and ambient temperatures, while embedded controllers adjust cooling mechanisms in real time to respond to dynamic thermal loads. Advanced systems may incorporate predictive models, machine learning algorithms, and feedback control to optimize energy consumption while maintaining temperature uniformity.

Hybrid thermal management approaches, combining passive and active cooling, have emerged as an effective solution to balance energy efficiency and thermal stability. For example, PCMs can absorb transient heat spikes, while airflow or liquid cooling can continuously remove heat to prevent PCM saturation. Such hybrid systems ensure better peak temperature control, reduced thermal gradients, and improved system reliability, especially under variable load profiles.

Overall, effective battery thermal management is crucial not only for maintaining performance and safety but also for enabling scalable and energy-efficient operation in diverse applications, from electric mobility platforms to grid-scale energy storage. These principles form the foundation for the design and implementation of the hybrid Air–PCM embedded thermal management system proposed in this project.

2.2 Air Cooling Techniques

Air cooling is one of the most widely used thermal management methods for lithium-ion battery systems due to its simplicity, low cost, and ease of implementation. In this approach, ambient or forced airflow is directed over the surface of battery cells or modules to remove heat generated during charge–discharge cycles. Air cooling can be implemented passively, relying on natural convection, or actively, using fans or blowers to increase airflow and enhance heat dissipation.

Passive air cooling is straightforward and requires minimal additional components, making it suitable for low-power or small-scale battery systems. However, its effectiveness is limited, particularly during high load or transient conditions, because it cannot provide sufficient heat removal when the thermal load exceeds the convective capacity of ambient air. Additionally, passive methods often result in uneven temperature distribution across the battery pack, leading to localized hot spots and accelerated cell aging.

Active air cooling overcomes some of these limitations by employing fans, blowers, or ducted channels to enhance convective heat transfer. Proper design of airflow paths, fan placement, and channel geometry is critical to achieving uniform temperature distribution. Experimental and computational studies have shown that optimized ducting and controlled airflow can significantly reduce peak temperatures and improve cell-to-cell thermal uniformity.

Despite these advantages, air cooling has inherent limitations. The heat removal capacity is restricted by the ambient air temperature and airflow rate, making it less effective in high-temperature environments or under high-power operation. Additionally, continuous operation of fans consumes electrical energy, which can reduce overall system efficiency, particularly in energy-constrained applications such as electric vehicles.

In summary, while air cooling techniques are simple and practical, their effectiveness is constrained by environmental conditions and load variability. These limitations highlight the need for hybrid solutions, such as integrating air cooling with Phase Change Materials (PCM), to achieve better thermal stability, energy efficiency, and uniform temperature control in modern battery energy storage systems.

Chapter 3 Methodology

3.1 Phase Change Material Based Cooling

Phase Change Material (PCM) based cooling is a passive thermal management technique that leverages the latent heat absorption property of materials during phase transition to regulate battery temperatures. When a battery generates heat during charge–discharge cycles, the PCM absorbs excess thermal energy by melting, effectively limiting the peak temperature rise of the cells and reducing temperature gradients across the battery module. This property makes PCM highly effective in smoothing transient thermal spikes and enhancing overall temperature uniformity.

The selection of an appropriate PCM is critical and depends on its melting temperature, latent heat capacity, thermal conductivity, and chemical stability. For lithium-ion batteries, PCMs with melting points aligned to the safe operating temperature range (typically 30–45°C for LFP cells) are preferred. To overcome the inherently low thermal conductivity of many PCMs, enhancements such as embedding aluminum fins, graphite sheets, or metal foams are commonly employed. These structures improve heat spreading, enabling more efficient energy absorption and distribution across the battery pack.

PCM-based cooling offers several advantages, including energy-free operation, silent performance, and the ability to handle short-term thermal transients effectively. Unlike active cooling systems, it does not require external power for heat absorption, making it an energy-efficient option for managing peak loads. Additionally, PCM can provide a safety buffer by delaying the onset of critical temperatures, allowing time for active cooling systems to respond if integrated in a hybrid architecture.

However, PCM systems also have limitations. Once the PCM reaches its fully melted state, its ability to absorb additional heat is reduced until it is "reset" by removing stored thermal energy. This makes PCM alone insufficient for continuous or high-power operations. Moreover, careful thermal design is necessary to ensure adequate contact between cells and PCM, minimize thermal resistance, and maintain structural integrity during repeated phase change cycles.

In modern BESS applications, PCM-based cooling is often combined with active cooling tech-

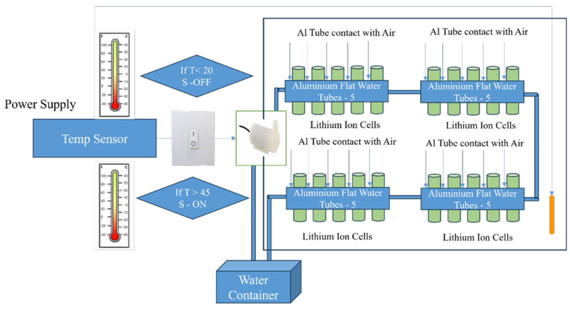

Figure 3.1: Methodology

niques such as forced airflow or liquid circulation to form a hybrid thermal management system. This integration allows PCM to absorb transient heat spikes while active cooling continuously removes accumulated heat, ensuring stable and uniform battery temperatures. Such hybrid systems exploit the strengths of both passive and active cooling, enhancing battery safety, performance, and operational efficiency.

3.2 Hybrid Thermal Management Systems

Hybrid systems combining PCM with active cooling have gained attention for their ability to handle both transient and steady-state heat loads. Experimental and simulation studies show that hybrid Air–PCM systems outperform standalone methods in terms of peak temperature reduction and energy efficiency. These systems, however, require intelligent control to coordinate passive and active components.A hybrid thermal management system is designed to achieve efficient and reliable temperature control by combining the advantages of both active and passive cooling techniques. In this project, the system integrates forced convection methods such as fans or coolant circulation with phase change materials (PCM) that absorb latent heat during temperature rise. When the system operates under high thermal loads, active cooling quickly removes excess heat, while the PCM stores surplus thermal energy by undergoing a phase transition. During low-load or cooling periods, the stored heat is gradually released, maintaining a uniform temperature distribution. This synergy minimizes thermal stress, reduces peak temperatures, and enhances overall energy efficiency. The hybrid approach not only improves system safety and operational stability but also extends the lifespan of components, making it an effective solution for advanced thermal management in battery energy storage systems and other high-power applications.The inclusion of phase change materials allows excess heat to be absorbed during peak operating conditions, improving overall thermal stability. By minimizing the continuous operation of active cooling components, the system enhances energy efficiency and reduces power consumption. As a result, the hybrid approach improves safety, reliability, and the operational lifespan of the system, making it well suited for high-power and energy storage applications.

3.3 Embedded and Intelligent Control Approaches

Recent literature highlights the role of embedded systems and control algorithms in advanced thermal management. Microcontroller-based platforms, combined with PID or model-based control, enable real-time adaptation to load variations. Emerging approaches also explore machine learning and predictive control to further optimize cooling performance and energy consumption.The embedded and intelligent control system is a core component of the hybrid thermal management system, enabling precise, real-time regulation of temperature under varying operating conditions. In this project, an embedded controller continuously acquires temperature data from strategically placed sensors and processes it using intelligent control logic. Based on the measured thermal load, the controller dynamically regulates active cooling elements such as fans or coolant pumps, ensuring that heat is dissipated efficiently during high-demand periods. At the same time, the control system coordinates with passive components, particularly phase change materials, which absorb excess heat during peak temperatures and release it gradually during cooling phases. This coordinated operation reduces sudden temperature fluctuations, minimizes thermal stress, and prevents overheating of critical components. Intelligent decision-making algorithms embedded within the controller optimize cooling performance by adjusting control parameters in real time, thereby reducing unnecessary energy consumption and improving overall system efficiency. Additionally, the embedded system enables features such as fault detection, safety protection, and data logging, which enhance operational reliability and allow performance analysis over extended periods. By integrating sensing, processing, and actuation into a single intelligent framework, the embedded control system ensures stable, adaptive, and energy-efficient thermal management. This approach significantly extends component lifespan, improves safety, and supports consistent performance, making it highly suitable for advanced applications such as battery energy storage systems and high-power electronic platforms.In addition to real-time control, the embedded and intelligent control framework supports predictive and adaptive thermal management by learning system behavior over time. Historical temperature data and operating patterns are analyzed to anticipate thermal spikes and initiate preventive cooling actions before critical limits are reached. This proactive control strategy further reduces response delay and enhances thermal safety. The embedded system also enables seamless integration with user interfaces and communication modules, allowing remote monitoring, parameter tuning, and system diagnostics. Such connectivity improves maintainability and scalability of the thermal management system. By combining intelligence, automation, and communication, the embedded control architecture transforms conventional thermal management into a smart, self-regulating system capable of meeting the demands of modern high-power and energy-intensive applications.

Chapter 4 Methodology

4.1 System Requirements and Design Goals

The primary goal of the proposed system is to maintain battery temperatures within safe limits while minimizing cooling energy. Key requirements include low cost, modularity, scalability, and reliability. The system targets a representative BESS module under controlled laboratory conditions.The methodology of this project begins with a clear definition of system requirements to ensure effective thermal regulation and intelligent control. The primary requirement is to maintain the operating temperature of critical components within safe limits under varying load conditions. This includes minimizing peak temperatures, reducing thermal gradients, and ensuring uniform heat distribution. Additional requirements involve achieving high energy efficiency, fast response to thermal variations, system reliability, and safe operation. The system must also support real-time monitoring, fault detection, and adaptability to different operating scenarios. Constraints such as compact size, low power consumption, cost-effectiveness, and ease of integration with existing hardware are also considered during requirement analysis.

Based on these requirements, the system design adopts a hybrid thermal management approach combining active and passive cooling techniques. Active cooling components such as fans or pumps are selected to provide rapid heat removal during high thermal loads, while passive elements like phase change materials are incorporated to absorb and store excess heat during peak conditions. Temperature sensors are strategically placed near heat-generating components to capture accurate real-time thermal data. An embedded controller forms the core of the system, interfacing with sensors and actuators to enable intelligent decision-making.

The design of the embedded and intelligent control system focuses on real-time data acquisition, processing, and control. Control algorithms are developed to dynamically regulate cooling intensity based on temperature thresholds and thermal trends. Safety mechanisms, including over-temperature protection and fault alerts, are integrated to ensure reliable operation. The system architecture is modular, allowing easy upgrades and scalability. Through careful requirement analysis and systematic design, the methodology ensures that the proposed system achieves efficient, adaptive, and reliable thermal management suitable for advanced energy storage and high-power electronic applications.In the next stage of system design, emphasis is placed on control strategy development and integration. The control logic is designed to operate in real time, continuously comparing sensed temperature values with predefined reference limits. Based on this comparison, the embedded controller determines the appropriate cooling action, such as increasing or decreasing fan speed or activating additional cooling paths. Threshold-based and adaptive control techniques are implemented to ensure fast response during sudden thermal rises while avoiding frequent switching that could reduce component life. This intelligent coordination between sensing and actuation ensures stable thermal regulation

Figure 4.1: Lithium-ion batteries pack (40 cells)

under both steady-state and transient conditions.

Another important aspect of the system design methodology is mechanical and thermal layout optimization. The placement of air channels, fans, PCM modules, and sensors is carefully planned to maximize heat transfer efficiency. Computational and analytical considerations are used to minimize airflow resistance and improve contact between heat sources and PCM modules. Thermal insulation and enclosure design are also considered to prevent heat leakage and external environmental effects. This optimized layout enhances the effectiveness of both active air cooling and passive PCM-based heat absorption, resulting in uniform temperature distribution throughout the system.

Finally, the system design includes testing, validation, and safety assurance. The hardware and control system are tested under different load and environmental conditions to evaluate thermal performance, response time, and energy efficiency. Fault conditions such as sensor failure, fan malfunction, or abnormal temperature rise are simulated to verify the robustness of safety mechanisms. Design iterations are carried out based on test results to improve system reliability and performance. Through systematic integration, optimization, and validation, the methodology ensures that the proposed hybrid thermal management system meets design objectives and operates safely and efficiently in real-world applications.

4.2 Hybrid Air–PCM Thermal Design

The thermal subsystem consists of PCM blocks placed in direct thermal contact with the battery module, enhanced using aluminum fins and graphite sheets to improve heat spreading. Ducted airflow driven by DC fans passes over the PCM and hot-spot regions to remove stored heat and reset the PCM during operation.The hybrid air–PCM (Phase Change Material) thermal management design is developed to provide efficient and reliable temperature control for systems subjected to varying thermal loads. This design combines the advantages of air-based active cooling with passive thermal energy storage using PCM, resulting in improved thermal stability and energy efficiency. The primary objective of the hybrid air–PCM system is to maintain component temperatures within safe operating limits while minimizing temperature fluctuations and reducing the dependence on continuous active cooling.

In the proposed design, air cooling is implemented using forced convection through strategically placed fans and airflow channels. These channels are designed to guide cool air uniformly across heat-generating components, ensuring effective heat removal during high-load operating conditions. The airflow path is optimized to reduce thermal resistance and prevent the formation of localized hot spots. Air cooling provides a rapid response to sudden temperature rises, making it suitable for dynamic operating environments.

Phase change materials are integrated into the system as passive thermal storage elements. The PCM is selected based on its melting temperature, high latent heat capacity, and thermal reliability to match the operating range of the system. PCM modules are placed in close thermal contact with heat sources, allowing them to absorb excess heat when temperatures exceed a predefined threshold. During this phase transition, the PCM stores thermal energy without a significant rise in temperature,

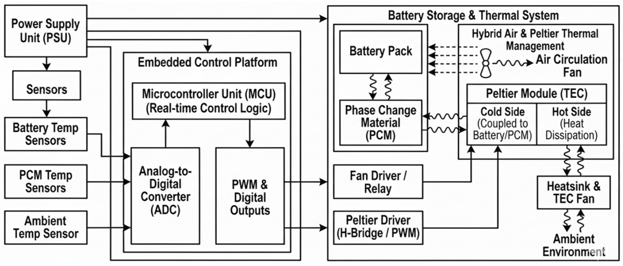

Figure 4.2: block diagram

thereby limiting peak temperatures. When the thermal load decreases, the stored heat is gradually released to the surrounding air, maintaining thermal balance.

The hybrid air–PCM design is coordinated through an embedded control system that monitors temperature in real time and regulates airflow accordingly. The controller adjusts fan speed based on thermal conditions, ensuring optimal interaction between active air cooling and passive PCM heat absorption. This synergistic design reduces energy consumption, enhances system safety, and extends component lifespan. Overall, the hybrid air–PCM thermal management design offers a compact, cost-effective, and intelligent solution for advanced battery energy storage systems and high-power electronic applications.

4.3 Embedded Hardware Architecture

The embedded control platform is built around a microcontroller (Arduino Uno or ESP32) interfaced with multiple temperature sensors (DS18B20) and ambient sensors (BME280). The controller drives PWM-controlled DC fans through MOSFET-based power stages. Data logging is implemented using an SD card, and a local HMI provides real-time system status.The embedded hardware architecture forms the backbone of the intelligent hybrid thermal management system, enabling real-time monitoring, control, and safe operation of the overall setup. The architecture is designed to ensure reliable data acquisition, efficient processing, and precise actuation while maintaining low power consumption and modularity. At the core of the hardware architecture is a microcontroller unit (MCU), which serves as the central processing and decision-making element. The MCU is selected based on parameters such as processing speed, input/output capability, low power operation, and support for real-time control applications. It executes the embedded control algorithms, manages sensor data, and controls the cooling actuators based on thermal conditions.

Temperature sensing is a critical part of the hardware design, and multiple temperature sensors are strategically placed near heat-generating components to capture accurate thermal data. These sensors may include digital temperature sensors or thermistors, interfaced to the MCU through analog-to-digital converters or digital communication protocols. Proper signal conditioning is incorporated to reduce noise and improve measurement accuracy. The use of multiple sensors allows spatial temperature monitoring, enabling the system to detect hot spots and thermal gradients effectively.

The actuation layer of the hardware architecture consists of cooling devices such as DC fans or air blowers, which are responsible for active air cooling. These actuators are driven through appropriate

Figure 4.3: ESP32 Embedded controller

driver circuits, including transistors or motor driver modules, to ensure safe and reliable operation. Pulse Width Modulation (PWM) techniques are employed to control fan speed, allowing fine-grained regulation of airflow based on real-time thermal demand. This approach improves energy efficiency and reduces unnecessary power consumption.

Power management is another key aspect of the embedded hardware architecture. A regulated power supply unit provides stable voltage levels to the MCU, sensors, and actuators. Protection components such as voltage regulators, fuses, and reverse polarity protection circuits are included to safeguard the system against electrical faults. The design ensures electrical isolation between low-power control circuitry and higher-power actuator circuits to enhance safety and reliability.

Additionally, the architecture supports communication and interface modules for system monitoring and diagnostics. Interfaces such as UART, I2C, or SPI enable data exchange between hardware components and external devices, facilitating parameter tuning and performance evaluation. Status indicators such as LEDs or displays provide visual feedback on system operation and fault conditions. Overall, the embedded hardware architecture is designed to be robust, scalable, and efficient, supporting intelligent thermal management while ensuring system safety, reliability, and long-term performance.

4.4 Control Algorithm and Firmware

Firmware implements sensor acquisition, signal filtering, and closed-loop control. Threshold-based logic with hysteresis ensures safe operation, while a PID controller modulates fan speed based on temperature error and gradients. Fault detection mechanisms handle sensor failures and over-temperature conditions.The control algorithm and firmware form the intelligence of the hybrid air–PCM thermal management system, enabling real-time decision-making, adaptive control, and reliable operation.

The firmware is developed and deployed on a microcontroller-based embedded platform, where it integrates temperature sensing, data processing, control logic, and actuation into a unified framework. The primary objective of the control algorithm is to maintain system temperature within predefined safe limits while optimizing energy consumption and ensuring smooth operation under varying thermal loads.

At the core of the control algorithm is a continuous temperature monitoring routine. Multiple temperature sensors placed near critical heat sources provide real-time thermal data to the microcontroller. The firmware periodically samples these sensor values using analog-to-digital conversion or digital communication interfaces. To improve measurement accuracy and robustness, digital filtering techniques such as moving average filtering are implemented to reduce noise and transient fluctuations. The processed temperature data is then compared against predefined threshold values corresponding to normal, warning, and critical temperature ranges.

Based on this comparison, the control logic dynamically regulates the active cooling components. Pulse Width Modulation (PWM) control is used to adjust fan speed proportionally to the thermal load. At low temperatures, the fans operate at minimal speed or remain inactive, allowing the phase change material to absorb heat passively. As temperature increases, the firmware incrementally increases the PWM duty cycle, providing higher airflow to enhance convective heat removal. This gradual control strategy prevents abrupt changes in airflow, reduces acoustic noise, and improves energy efficiency. During peak thermal conditions, the algorithm ensures maximum cooling while maintaining system stability.

The firmware also coordinates the interaction between active air cooling and passive PCM behavior. When temperatures approach the PCM melting range, the algorithm prioritizes passive heat absorption by limiting excessive fan operation, thereby utilizing the latent heat storage capability of the PCM. Once the PCM becomes saturated or when temperatures continue to rise, active cooling is intensified to prevent overheating. This intelligent coordination enhances thermal stability and reduces dependence on continuous active cooling.

Safety and fault-handling mechanisms are integral parts of the control firmware. The algorithm continuously checks for abnormal conditions such as sensor failure, communication errors, or fan malfunction. In the event of a fault, the system triggers protective actions, including activating maximum cooling, generating alerts, or safely shutting down non-critical loads. Status indicators and diagnostic messages are used to provide real-time feedback for monitoring and maintenance.

Additionally, the firmware supports data logging and communication interfaces for performance analysis and system tuning. Operational parameters such as temperature trends, fan duty cycles, and fault events are stored or transmitted for evaluation. The modular structure of the firmware allows easy updates and scalability for future enhancements. Overall, the control algorithm and firmware ensure adaptive, efficient, and safe thermal management, enabling the hybrid air–PCM system to perform reliably in advanced battery energy storage and high-power electronic applications.

4.5 Experimental Setup and Testing Procedure

A bench-scale prototype is assembled using heater elements to emulate battery heat generation. Step and cyclic load profiles are applied under varying ambient conditions. Performance metrics such as peak temperature, temperature uniformity, cool-down time, and cooling energy are recorded for analysis.The experimental setup and testing process are designed to evaluate the performance, reliability, and effectiveness of the proposed hybrid air–PCM thermal management system under controlled and repeatable conditions. The experimental setup consists of the integrated thermal management assembly, embedded control hardware, sensing elements, and load simulation components arranged to replicate real operating environments. Heat-generating elements, such as electrical heaters or power electronic modules, are used to simulate thermal loads equivalent to those encountered in practical applications like battery energy storage systems or high-power electronics.

The hybrid air–PCM system is assembled with phase change material modules placed in close thermal contact with the heat source to ensure efficient heat absorption. Forced air cooling is provided through DC fans positioned along optimized airflow channels. Temperature sensors are installed at critical locations, including near the heat source, within the PCM region, and at the air inlet and outlet, to capture spatial temperature variations. These sensors are interfaced with the embedded controller, enabling real-time temperature monitoring and control during experiments. A regulated power supply is used to provide stable power to the control circuitry, sensors, and actuators, ensuring consistent operating conditions throughout the tests.

The testing process begins with baseline experiments to evaluate system behavior without active control. In this phase, only passive PCM cooling is employed to observe its heat absorption capability and temperature stabilization performance. The results from this test provide reference data for comparison with active and hybrid cooling modes. Subsequently, tests are conducted using only active air cooling to analyze the effectiveness of forced convection in heat removal and its impact on energy consumption. These individual mode tests help in understanding the contribution of each cooling technique.

In the hybrid testing phase, both air cooling and PCM are operated under embedded intelligent control. The system is subjected to varying thermal loads, including steady-state, step changes, and peak load conditions. The control algorithm dynamically adjusts fan speed based on real-time temperature feedback, while the PCM absorbs excess heat during peak temperatures. Key performance metrics such as peak temperature reduction, temperature uniformity, response time, and power consumption are recorded and analyzed.

Additional tests are performed to evaluate system robustness and safety. Fault scenarios such as sensor disconnection, fan failure, and sudden load surges are introduced to verify the effectiveness of protection mechanisms. Long-duration tests are conducted to assess thermal cycling behavior and system stability over extended operation. The experimental data is logged and analyzed to validate design objectives and demonstrate the advantages of the hybrid air–PCM thermal management system. Overall, the experimental setup and testing process confirm the system's ability to deliver efficient, reliable, and intelligent thermal control under real-world operating conditions.

Chapter 5 Results and Discussion

5.1 Temperature Performance Analysis

Experimental results show that the hybrid Air–PCM system significantly reduces peak temperatures compared to air-only cooling. The PCM absorbs transient heat loads, limiting temperature spikes during high-power operationThe temperature performance and analysis of the hybrid air–PCM thermal management system focus on evaluating its ability to regulate heat effectively under varying operating conditions. Temperature data obtained from multiple sensors placed near the heat source, PCM region, and airflow path are analyzed to assess thermal stability, peak temperature reduction, and uniformity. The system performance is examined under passive, active, and hybrid operating modes to highlight the advantages of integrated thermal management.

During passive PCM operation, the temperature rise is initially rapid until the PCM reaches its phase transition range. Once the melting point is reached, the PCM absorbs a significant amount of heat as latent energy, resulting in a noticeable reduction in the rate of temperature increase. This behavior demonstrates the PCM's effectiveness in limiting peak temperatures during short-duration or moderate thermal loads. However, prolonged operation leads to PCM saturation, after which temperature begins to rise steadily, indicating the limitation of passive cooling when used alone.

In active air-cooling mode, the system shows a faster response to temperature rise due to forced convection. Increased airflow effectively removes heat, leading to lower steady-state temperatures compared to passive cooling. However, continuous fan operation results in higher energy consumption and increased temperature fluctuations during load variations. This highlights the trade-off between rapid heat removal and energy efficiency in purely active cooling systems.

The hybrid air–PCM mode exhibits the best overall thermal performance. Peak temperatures are significantly reduced compared to both passive and active modes, and temperature variations are smoother and more uniform. The PCM absorbs excess heat during peak conditions, while air cooling maintains thermal balance and assists in releasing stored heat during low-load periods. This coordinated operation minimizes hot spots, reduces thermal stress, and improves temperature uniformity across the system. Overall, the temperature analysis confirms that the hybrid approach provides superior thermal stability, enhanced safety, and improved energy efficiency, making it highly suitable for advanced battery energy storage and high-power electronic applications..

5.2 Temperature Uniformity and ΔT Evaluation

The system demonstrates improved temperature uniformity across the battery module, with measured temperature differences (ΔT) remaining within acceptable limits. This uniformity is critical for balanced aging and enhanced battery reliability.Temperature uniformity and delta (T) analysis are critical metrics for evaluating the effectiveness of the hybrid air–PCM thermal management system, as they directly influence system reliability, safety, and component lifespan. Temperature uniformity refers to the ability of the system to maintain similar temperature levels across different regions, while delta analysis focuses on the temperature difference between the hottest and coolest points within the system. A lower T indicates reduced thermal gradients and improved heat distribution, which are essential for preventing localized overheating and thermal stress.

In passive PCM operation, temperature uniformity improves significantly once the PCM enters its phase change region. During this stage, the PCM absorbs excess heat at nearly constant temperature, which helps reduce temperature differences between the heat source and surrounding regions. However, before melting and after saturation, temperature gradients increase, resulting in a higher T across the system. This behavior highlights the time-dependent effectiveness of PCM when used alone and its limitation under prolonged or high thermal loads.

Active air cooling improves temperature uniformity by promoting convective heat transfer and distributing heat away from localized hot spots. Increased airflow reduces the temperature difference between components, especially during steady-state operation. However, non-uniform airflow distribution and varying heat generation rates can still lead to localized temperature variations. Sudden changes in load may also cause transient temperature deltas due to the response delay of the cooling system.

The hybrid air–PCM system demonstrates superior temperature uniformity and the lowest T values among all operating modes. The PCM absorbs localized heat peaks, while forced air circulation evenly distributes thermal energy across the system. This combined mechanism significantly reduces spatial temperature differences and smooths transient thermal behavior. Delta analysis shows a substantial reduction in maximum temperature difference during peak load conditions compared to passive or active cooling alone. Furthermore, the intelligent control of fan speed ensures balanced airflow, preventing overcooling in certain regions and undercooling in others.

Overall, the temperature uniformity and delta analysis confirm that the hybrid air–PCM thermal management system effectively minimizes thermal gradients and maintains consistent temperature distribution. This improved uniformity reduces mechanical stress, enhances system safety, and extends component lifespan. The results validate the suitability of the proposed system for high-power and energy storage applications where maintaining low temperature differentials is crucial for reliable and long-term operation.

5.3 Cooling Energy and Efficiency

Analysis of fan power consumption indicates that the hybrid system achieves lower overall cooling energy for the same thermal mission. PCM buffering reduces the need for continuous high-speed fan operation, improving energy efficiency.Cooling energy and efficiency analysis is essential to evaluate the practicality and sustainability of the proposed hybrid air–PCM thermal management system. Cooling energy refers to the electrical power consumed by active cooling components such as fans or blowers, while efficiency indicates how effectively this energy is utilized to control temperature and reduce thermal stress. An efficient system should achieve the desired temperature regulation with minimal energy consumption, especially for long-duration and high-power applications.

In purely active air-cooling mode, cooling energy consumption is relatively high due to continuous fan operation. As thermal load increases, fan speed must be increased to maintain safe temperatures, leading to higher power usage. Although active cooling provides rapid heat removal, it often operates even during moderate thermal conditions, resulting in unnecessary energy expenditure. This mode also shows reduced efficiency during transient conditions, as sudden fan speed changes are required to respond to load variations.

In contrast, the hybrid air–PCM system demonstrates significantly improved cooling energy efficiency. During low to moderate thermal loads, the phase change material absorbs excess heat passively without requiring electrical energy. This reduces the need for continuous fan operation and allows the system to operate at lower fan speeds for extended periods. As a result, overall cooling energy consumption is substantially reduced compared to active-only cooling. During peak thermal conditions, active air cooling is selectively intensified, ensuring safety while still benefiting from the thermal buffering provided by the PCM.

Efficiency analysis shows that the hybrid approach achieves a better balance between thermal performance and energy usage. For the same level of peak temperature reduction, the hybrid system consumes less electrical power than the active cooling system alone. Intelligent control algorithms further enhance efficiency by dynamically adjusting fan speed based on real-time temperature data, avoiding overcooling and frequent on–off cycling. This adaptive operation minimizes power losses and improves system responsiveness.

Additionally, reduced cooling energy consumption contributes to lower operational costs and extended system lifespan. Lower fan usage decreases mechanical wear, reduces noise, and improves overall system reliability. The improved efficiency also makes the hybrid system more suitable for energy-sensitive applications such as battery energy storage systems, electric vehicles, and portable electronics. Overall, the cooling energy and efficiency analysis confirms that the hybrid air–PCM thermal management system delivers effective temperature control while achieving significant energy savings, validating its advantage over conventional cooling methods.

5.4 Dynamic Response and Stability

The embedded control algorithm provides stable and responsive cooling behavior. The use of hysteresis and PID control prevents oscillations and ensures smooth transitions between operating states. Dynamic response and stability are critical performance metrics for evaluating the effectiveness of the hybrid air–PCM thermal management system. Dynamic response refers to how quickly the system reacts to changes in thermal load, such as sudden increases in heat generation, while stability measures the system's ability to maintain consistent temperatures without oscillations or overshoot after a disturbance. An effective thermal management system must exhibit fast response times to prevent overheating while maintaining stable operation to avoid thermal stress and energy inefficiency.

In the hybrid system, the dynamic response is enhanced by the combination of active and passive cooling mechanisms. Active air cooling, driven by DC fans and controlled through Pulse Width Modulation (PWM), responds immediately to rising temperatures by increasing airflow. This rapid action prevents sharp temperature spikes and protects critical components. Simultaneously, the PCM provides thermal buffering by absorbing excess heat during transient peaks, delaying temperature rise and smoothing the system's thermal response. The synergy between the immediate effect of active cooling and the latent heat absorption of PCM ensures a more gradual and controlled temperature change, improving both response time and operational safety.

Stability analysis shows that the hybrid system maintains consistent temperature levels under varying loads and environmental conditions. The embedded control algorithms continuously monitor temperature data and adjust fan speed in real time, preventing overshoot or oscillations that can occur with purely active systems. The PCM contributes to damping rapid temperature fluctuations, acting as a thermal capacitor that reduces the amplitude of transient variations. This coordinated approach ensures that after a thermal disturbance, the system returns to equilibrium quickly and remains within safe operating limits.

Testing under step load changes and periodic thermal cycles confirms that the hybrid system exhibits both fast dynamic response and high stability. The maximum temperature overshoot is minimized, and temperature oscillations are damped effectively compared to active-only or passive-only cooling modes. Furthermore, the intelligent control allows adaptive adjustment based on system behavior, maintaining stability even under unpredictable or varying loads.

Overall, the hybrid air–PCM thermal management system demonstrates superior dynamic response and thermal stability. It effectively balances rapid heat removal with thermal buffering, minimizes temperature overshoot, and ensures uniform, stable operation. This makes it highly suitable for high-power applications, energy storage systems, and electronic devices where both fast response and stable long-term thermal performance are essential.

5.5 Comparison with Baseline Systems

When compared to an air-only baseline, the proposed hybrid system shows clear advantages in peak temperature reduction, thermal uniformity, and energy efficiency. These results validate the effectiveness of combining PCM with controlled airflow and embedded intelligence.

The performance of the hybrid air–PCM thermal management system is evaluated by comparing it with a baseline system that employs conventional cooling methods, typically consisting of either active air cooling alone or passive PCM-only cooling. This comparison highlights the advantages of the hybrid approach in terms of temperature control, energy efficiency, thermal stability, and overall system reliability.

In terms of temperature regulation, the baseline active cooling system responds quickly to thermal loads but often exhibits higher temperature fluctuations and localized hot spots due to uneven airflow distribution. Passive PCM-only systems, on the other hand, reduce peak temperatures moderately through latent heat absorption but fail to manage prolonged or high thermal loads effectively, leading to gradual temperature rise after PCM saturation. The hybrid system combines both methods, providing faster heat removal through active airflow while the PCM absorbs transient heat peaks. As a result, the hybrid system maintains lower peak temperatures and smoother temperature profiles compared to both baseline modes.

Regarding temperature uniformity, the baseline active system shows moderate uniformity depending on airflow effectiveness, while PCM-only systems offer better uniformity during the phase change period but are limited under extended operation. The hybrid system achieves the best temperature uniformity by leveraging the synergistic effect of airflow distribution and latent heat absorption. Delta analysis (T between hottest and coolest points) confirms that the hybrid system reduces thermal gradients significantly compared to the baseline, minimizing thermal stress on components.

From an energy efficiency perspective, the baseline active system consumes higher electrical power due to continuous fan operation, whereas PCM-only systems have negligible energy usage but limited cooling capacity. The hybrid system reduces energy consumption by utilizing PCM to handle moderate heat loads passively while activating fans only when necessary. Intelligent control algorithms further optimize fan operation, enhancing overall efficiency and reducing operational costs relative to the baseline system.

In terms of dynamic response and stability, the baseline systems exhibit limitations. Active cooling responds quickly but may overshoot and oscillate, while passive cooling is slow and cannot adapt to rapid load changes. The hybrid system demonstrates both fast dynamic response and stable operation, with minimal overshoot and damped temperature fluctuations, owing to the combined effect of active airflow and PCM buffering.

Overall, the hybrid air–PCM thermal management system outperforms baseline systems by providing superior temperature control, improved uniformity, enhanced energy efficiency, and reliable dynamic stability. This makes it a highly effective solution for high-power electronics, battery energy storage, and applications requiring precise and intelligent thermal management.In terms of dynamic response and stability, the baseline systems exhibit limitations. Active cooling responds quickly but may overshoot and oscillate, while passive cooling is slow and cannot adapt to rapid load changes. The hybrid system demonstrates both fast dynamic response and stable operation, with minimal overshoot and damped temperature fluctuations, owing to the combined effect of active airflow and PCM buffering. Furthermore, the hybrid system supports long-term thermal cycling with consistent performance, whereas baseline systems may degrade under repeated thermal stress.

Overall, the hybrid air–PCM thermal management system outperforms baseline systems by providing superior temperature control, improved uniformity, enhanced energy efficiency, and reliable dynamic stability. This makes it a highly effective solution for high-power electronics, battery energy storage, and applications requiring precise, intelligent, and energy-optimized thermal management, ensuring safe and long-lasting operation.

Chapter 6 Conclusion

6.1 Summary of Work

This project presented the design and implementation of a real-time embedded control platform for hybrid Air–PCM thermal management in battery energy storage systems. The system integrates passive and active cooling with intelligent control to address key thermal challenges.The present work focuses on the design, development, and evaluation of a hybrid air–PCM thermal management system with embedded intelligent control, aimed at efficiently managing heat in high-power electronic systems and battery energy storage applications. The primary objective of the study is to maintain component temperatures within safe operating limits, improve energy efficiency, and ensure long-term reliability by combining active and passive cooling techniques. The hybrid approach leverages the rapid heat dissipation capability of forced air cooling with the latent heat absorption properties of phase change materials (PCM), resulting in a more stable and energy-efficient thermal management solution compared to conventional methods.

The methodology of the work began with a detailed system requirement analysis, defining critical parameters such as temperature limits, energy consumption, response time, and uniformity. Based on these requirements, the system architecture was designed to integrate air channels, PCM modules, temperature sensors, and an embedded control platform. The embedded hardware includes a microcontroller, sensor interface, actuator drivers, and power regulation units, ensuring accurate data acquisition, real-time processing, and precise actuation. Control firmware was developed to implement intelligent algorithms that dynamically adjust fan speed and coordinate passive PCM operation based on real-time temperature readings. Safety features, fault detection, and data logging were incorporated to ensure robust operation and long-term reliability.

The experimental setup was developed to simulate real operating conditions using heat-generating elements as thermal loads. Temperature sensors were placed near critical components, PCM modules, and airflow paths to capture spatial thermal distribution. Experiments were conducted under passive, active, and hybrid operating modes to evaluate peak temperature reduction, temperature uniformity, cooling energy consumption, dynamic response, and stability. Data from these tests were analyzed to compare the performance of the hybrid system with conventional baseline systems, highlighting improvements in thermal regulation, reduced temperature gradients, faster response to transient loads, and enhanced energy efficiency.

The results demonstrate that the hybrid air–PCM system significantly outperforms baseline systems. Peak temperatures were reduced, temperature uniformity improved, and thermal gradients minimized, thereby reducing component stress. Intelligent control ensured rapid dynamic response and stable operation under varying loads, while PCM integration reduced reliance on continuous active cooling, lowering energy consumption. The system also exhibited robust performance under

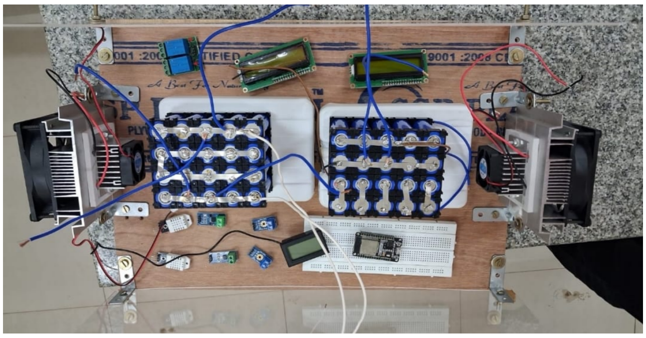

Figure 6.1: working model

fault conditions and long-duration operation, validating the reliability and effectiveness of the design.

Overall, this work successfully demonstrates the feasibility and advantages of a hybrid air–PCM thermal management system with embedded intelligent control. By combining passive and active cooling strategies with real-time adaptive control, the system achieves efficient, stable, and energy-optimized thermal management, making it highly suitable for advanced battery energy storage systems, high-power electronics, and other applications requiring reliable thermal regulation and long-term operational safety.

6.2 Key Achievements

The developed prototype successfully demonstrated reduced peak temperatures, improved thermal uniformity, and energy-efficient cooling. The embedded platform proved to be reliable, modular, and suitable for laboratory-scale BESS evaluation.The hybrid air–PCM thermal management project with embedded intelligent control has achieved several significant outcomes that demonstrate both innovation and practical effectiveness in advanced thermal management applications. One of the primary achievements is the successful integration of active air cooling with passive PCM-based heat absorption, resulting in a hybrid system capable of maintaining component temperatures within safe operational limits under varying thermal loads. This integration ensures a balanced approach where active cooling responds rapidly to sudden heat spikes while PCM provides latent heat storage, reducing peak temperatures and improving thermal stability.

Another key achievement is the development and implementation of an embedded intelligent control system. The control firmware enables real-time monitoring of multiple temperature sensors, dynamic adjustment of fan speed through PWM control, and coordinated operation with PCM modules. This intelligent control ensures adaptive cooling, minimizes energy consumption, and enhances the system's response to transient thermal events. Additionally, fault detection and safety mechanisms embedded in the firmware improve operational reliability, making the system robust against sensor failures, fan malfunctions, or abnormal thermal conditions.

The project also achieved excellent temperature uniformity and reduced thermal gradients, as demonstrated through delta (T) analysis. By minimizing temperature differences across components, the system effectively reduces thermal stress, enhances reliability, and prolongs component life. Experimental testing further confirmed improved energy efficiency, with the hybrid system consuming less electrical power than conventional active-only cooling systems, thanks to the passive heat absorption capability of the PCM and intelligent fan operation.

Finally, the system demonstrated superior dynamic response and stability, responding quickly to sudden load changes while maintaining steady-state temperatures without overshoot or oscillation. Long-duration testing verified consistent performance under thermal cycling, validating the system's reliability and applicability in real-world conditions.

Overall, the project successfully combines advanced thermal management techniques, intelligent control, and energy-efficient operation, offering a robust and practical solution for high-power electronics, battery energy storage systems, and other applications requiring precise and reliable temperature regulation.

6.3 Limitations of the Present Work

The current implementation is limited to bench-scale testing and simplified load profiles. Long-term aging effects and large-scale deployment were not addressed within the project scope.While the hybrid air–PCM thermal management system demonstrates significant improvements in temperature regulation, energy efficiency, and dynamic stability, certain limitations remain in the present work. One of the primary limitations is the capacity and saturation of the phase change material (PCM). During prolonged high thermal loads, the PCM may become fully melted and saturated, reducing its ability to absorb additional heat. In such cases, the system relies primarily on active air cooling, which may lead to higher energy consumption and slightly reduced thermal performance compared to ideal conditions.

Another limitation is the dependence on fan efficiency and airflow distribution. Although the embedded control adjusts fan speed dynamically, non-uniform airflow or mechanical limitations of fans can create localized hot spots in certain regions. These variations, though minimized, may affect overall temperature uniformity and slightly reduce the effectiveness of hybrid cooling under extreme load conditions.

The experimental setup and scale of the current system also present limitations. The work has been primarily tested on a laboratory-scale prototype with simulated thermal loads. While the results are promising, performance under larger-scale systems or real-world operational environments may differ due to variations in heat generation, airflow resistance, and external environmental factors such as ambient temperature and humidity.

Additionally, the embedded control algorithms used in the present work are primarily threshold- and rule-based. While effective, they may not fully exploit predictive or adaptive control strategies that could further enhance efficiency and response under highly dynamic or unpredictable thermal loads.

Lastly, material and integration constraints may limit the flexibility of the system in certain compact or space-constrained applications. PCM placement, airflow channel design, and actuator integration require careful design, which may not be easily adaptable for all geometries or high-density electronic assemblies.

Overall, while the present work demonstrates significant advantages, these limitations highlight areas for future improvement, including enhanced PCM capacity, optimized airflow design, predictive control strategies, and scalability for real-world applications.

6.4 Future Enhancements

Future work may include advanced control strategies such as model predictive control, machine learning-based thermal prediction, and real-time optimization. Integration with IoT and cloud platforms for remote monitoring and analytics is also a promising directions.

The hybrid air–PCM thermal management system presents several opportunities for future enhancement to further improve performance, efficiency, and applicability. One key area of enhancement is the integration of advanced phase change materials (PCMs) with higher latent heat capacity and tailored melting points. Using PCMs with faster thermal response or multiple phase change layers could increase heat absorption during peak loads and extend effective cooling duration, reducing reliance on active cooling.

Another potential improvement is the implementation of predictive and adaptive control algorithms. Machine learning or model predictive control (MPC) techniques could be incorporated into the embedded firmware to anticipate thermal spikes based on historical data and operating patterns. This would enable proactive cooling adjustments, reducing overshoot, improving energy efficiency, and enhancing dynamic response under variable or unpredictable thermal loads.

Enhanced airflow and mechanical design can also improve system performance. Optimized fan placement, duct design, or the use of micro-channel airflow systems could provide more uniform cooling, minimize hot spots, and reduce energy consumption. Additionally, using variable-speed or more efficient fans could further enhance energy efficiency and noise reduction.

Future work could also focus on scalability and integration for real-world applications. The system could be adapted for larger battery packs, high-power electronics, or compact devices, with design optimization to fit various form factors. Incorporating wireless monitoring and IoT-based connectivity would allow remote supervision, predictive maintenance, and automated alerts for temperature anomalies.

Finally, multi-modal hybrid cooling approaches can be explored, combining air, PCM, and other cooling methods such as liquid cooling or thermoelectric modules. This would further expand the system's thermal management capacity and adaptability under extreme or highly variable conditions.

Overall, these enhancements would make the hybrid air–PCM thermal management system more intelligent, energy-efficient, robust, and versatile, enabling it to meet the demands of next-generation high-power electronics, battery energy storage systems, and other advanced applications.

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